

Incorrect Data Link Transmission from Diesel Control Unit to Vessel

Symptoms

- The DCU410 or remote panel does **not** indicate faults because it does **not** have power +24-VDC.
- The DCU410 unit and remote panel LEDs illuminated.
- The ECM has active faults

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

The SAE J1939 data link delivers alarm information, generated by the ECM, to the customer interface box. The customer interface box transfers alarm information to the DCU410 and remote panel.

This tree addresses the DCU410 power supply and return.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box wiring.	
	STEP 1A. Check the DCU410 unit display for indication of faults.	
	STEP 1A-1. Check the DCU410 power supply wire for voltage +24-VDC.	
	STEP 1B. Check remote panel wiring.	

STEPS		SPECIFICATIONS
	STEP 1B-1. Check the remote panel power supply wire for voltage +24-VDC.	
STEP 2.	Check customer interface box wiring.	
	STEP 2A. Check the DCU410 controller area network open supply and return wires at the DCU410 and X4 connection for an open.	
	STEP 2B. Check the DCU410 controller area network open supply and return wire at the DCU410 unit and X4 connection for a wire-to-wire short.	
	STEP 2C. Check the DCU410 controller area network open supply wire at the DCU410 unit and X4 connection for a short to ground.	
	STEP 2D. Check the SAE J1939 data link supply and return wires at the DCU410 unit, X4 connection, and C3 connector for an open.	
	STEP 2D-1. Check the SAE J1939 data link supply and return wires at the DCU410 unit, X4 connection, and C3 connector for a wire-to-wire short.	
	STEP 2D-2. Check the SAE J1939 data link supply wire at the DCU410 unit, X4 connection, and C3 connector for a short to ground.	
STEP 3.	Check the OEM Wiring Harness	
	STEP 3A. Check the SAE J1939 data link signal and return wires for an open.	

STEPS		SPECIFICATIONS
	STEP 3A-1. Check the SAE J1939 data link signal and return wires for a wire-to-wire short.	
	STEP 3A-2. Check the SAE J1939 data link signal wire for a short to ground.	

STEP 1. Check the customer interface box wiring.

STEP 1A. Check the DCU410 unit display for faults.

Conditions :

- Locate the DCU410 unit display.

Action	Specification/Repair	Next Step
Check the DCU410 unit display for indication of faults.	The DCU410 unit indicates fault(s)? YES	1B
	The DCU410 unit indicates fault(s)? NO	1A-1

STEP 1A-1. Check the DCU410 power supply wire for voltage +24-VDC.

Conditions :

- Open the customer interface box.

Action	Specification/Repair	Next Step
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Check the the voltage at the battery 1 voltage (switched power) at the DCU410 unit. - Place one test lead on the battery 1 voltage (switched power) supply where at the DCU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? YES Repair : Check the batteries. Refer to the OEM service manual.	Repair complete
	Less than +24-VDC? NO	1B

STEP 1B. Check the remote panel wiring.

Conditions : Locate the remote panel display.		
Action	Specification/Repair	Next Step
Check the remote panel display for an indication of faults.	Does the remote panel unit indicate fault(s)? YES	Repair complete
	Does the remote panel unit indicate fault(s)? NO	1B-1

STEP 1B-1. Check the remote panel power supply wire for voltage +24-VDC.

Conditions : Open the remote panel.		
Action	Specification/Repair	Next Step

Check the the voltage at the battery 1 voltage (switched power) at the DCU410 unit. • Place one test lead on the battery 1 voltage (switched power) supply wire at the DCU410 unit. • Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than +24-VDC? YES Repair : Check the batteries. Refer to the OEM service manual.	Repair complete
	Less than +24-VDC? NO	2A

STEP 2. Check the customer interface box wiring.

STEP 2A. Check the DCU410 controller area network open supply and return wires at the DCU410 unit and X4 connection for an open.

Conditions : • Open the customer interface box. • Disconnect the DCU410 controller area network open supply and return wires at the DCU410 and X4 connection.		
Action	Specification/Repair	Next Step

Check the supply and return wires for an open. - Place one test lead on the DCU410 controller area network open supply wire at the DCU410 unit. - Place the other test lead on the DCU410 network open supply wire at the X4 connection. - Place one test lead on the DCU410 controller area network open return wire at the DCU410 unit. - Place the other test lead on the DCU410 network open return wire at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 2B. Check the DCU410 controller area network open supply and return wire at the DCU410 unit and X4 connection for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the DCU410 controller area network open supply wire at the DCU410 unit and X4 connection.

Action	Specification/Repair	Next Step

Check the supply and return wires for a wire-to-wire short. - Place one test lead on the DCU410 controller area network open supply wire at the DCU410 unit. - Place the other test lead on all other wires at the DCU410 unit. - Place one test lead on the DCU410 controller area network open return wire at the X4 connection. - Place the other test lead on all other wires at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	2C

STEP 2C. Check the DCU410 controller area network open supply wire at the DCU410 unit and X4 connection for a short to ground.

Conditions : - Open the customer interface box. - Disconnect the DCU410 controller area network open supply wire at the DCU410 unit and X4 connection.		
Action	Specification/Repair	Next Step

Check the supply wire for a short to ground. - Place one test lead on the DCU410 controller area network open supply wire at the DCU410 unit. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html) Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	2D

STEP 2D. Check the SAE J1939 data link supply and return wires at the DCU410 unit, X4 connection, and C3 connector for an open.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires at the DCU410 unit and X4 connection.
- Disconnect the C3 connector.

Action	Specification/Repair	Next Step
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Check the SAE J1939 data link supply and return wires for an open. - Place one test lead on the SAE J1939 data link supply wire at the DCU410 unit. - Place the other test lead on the SAE J1939 data link supply wire at the X4 connection. - Place the other test lead on the SAE J1939 data link supply pin at the C3 connector. - Place one test lead on the SAE J1939 data link return wire at the DCU410 unit. - Place the other test lead on the SAE J1939 data link return wire at the X4 connection. - Place the other test lead on the SAE J1939 data link return pin at the C3 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	2D-1
	Less than 10 ohms? NO Repair : Replace the wire. Refer to Procedure 015-023 (Customer Interface Box) in Section 15. (/qs3/pubsys2/xml/en/procedures/116/116-015-023.html)	Repair complete

STEP 2D-1. Check the SAE J1939 data link supply and return wires at the DCU410 unit, X4 connection, and C3 connector for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link supply and return wires at the DCU410 unit and X4 connection.
- Disconnect the C3 connector.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply and return wires at the DCU410 unit, X4 connection, and C3 connector for a wire-to-wire short.

- Place one test lead on the SAE J1939 data link supply wire at the DCU410 unit.
- Place the other test lead on all other wires at the X4 connection.
- Place one test lead on the SAE J1939 data link return wire at the DCU410 unit.
- Place the other test lead on all other wires at the DCU410 unit.
- Place one test lead on the SAE J1939 data link supply wire at the X4 connection.
- Place the other test lead on all other wires at the X4 connection.
- Place one test lead on the SAE J1939 data link return wire at the X4 connection.
- Place the other test lead on all other wires at the X4 connection.
- Place one test lead on the SAE J1939 data link supply pin at the C3 connector.
- Place the other test lead on all other pins at the C3 connector.

Less than 10 ohms?

YES

Repair :

Replace the DCU410 unit.
Contact a Cummins®
Authorized Repair Location.

Repair complete

- Place one test lead on the SAE J1939 data link return pin at the C3 connector. - Place the other test lead on all other pins at the C3 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? NO	2D-2
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STEP 2D-2. Check the SAE J1939 data link supply wire at the DCU410 unit, X4 connection, and C3 connector for a short to ground.		
Conditions : - Open the customer interface box. - Disconnect the SAE J1939 data link supply wire at the DCU410 unit and X4 connection. - Disconnect the C3 connector.		
Action	Specification/Repair	Next Step

Check the SAE J1939 data link supply wire at the DCU410 unit, X4 connection, and C3 connector for a short to ground. - Place one test lead on the SAE J1939 data link supply wire at the DCU410 unit. - Place the other test lead on panel ground. - Place one test lead on the SAE J1939 data link supply wire at the X4 unit. - Place the other test lead on panel ground. - Place one test lead on the SAE J1939 data link supply pin at the C3 connector. - Place the other test lead on panel ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the DCU410 unit. Contact a Cummins® Authorized Repair Location.	Repair complete
	Less than 10 ohms? NO	3A

STEP 3. Check the OEM wiring harness.

STEP 3A. Check the SAE J1939 data link signal and return wires for an open.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link.
- Disconnect the SAE J1939 data link signal and return wires at the X4 connection.

Action	Specification/Repair	Next Step
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Check the SAE J1939 data link signal and return wires for an open. - Place one test lead on the SAE J1939 data link signal wire at the X4 connection. - Place the other test lead on the SAE J1939 data link signal pin at the SAE J1939 data link service port connector. - Place one test lead on the SAE J1939 data link return wire at the X4 connection. - Place the other test lead on the SAE J1939 data link return pin at the SAE J1939 data link service port connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	3A-1
	Less than 10 ohms? NO Repair : Replace the wire. Refer to the OEM service manual.	Repair complete

STEP 3A-1. Check the SAE J1939 data link signal and return wires for a wire-to-wire short.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link.
- Disconnect the SAE J1939 data link signal and return wires at the X4 connection.

Action	Specification/Repair	Next Step
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Check the SAE J1939 data link signal and return wires for a wire-to-wire short. • Place one test lead on the SAE J1939 data link signal wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. • Place one test lead on the SAE J1939 data link return wire at the X4 connection. • Place the other test lead on all other wires at the X4 connection. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO	3A-2

STEP 3A-2. Check the SAE J1939 data link signal wire for a short to ground.

Conditions :

- Open the customer interface box.
- Disconnect the SAE J1939 data link.
- Disconnect the SAE J1939 data link signal wire at the X4 connection.

Action	Specification/Repair	Next Step
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Check the SAE J1939 data link signal wire for a short to ground. • Place one test lead on the SAE J1939 data link signal wire at the X4 connection. • Place the other test lead on engine ground. • Place one test lead on the SAE J1939 data link return wire at the X4 connection. • Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO Repair : Replace the SAE J1939 data link service port connector. Contact a Cummins® Authorized Repair Location.	Contact a Cummins® Authorized Repair Location.